

Slide 1

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Good morning, everyone. My name is Jonathan Norton, and welcome to my CS 499 Capstone code review presentation. Today I will demonstrate my growth throughout the Computer Science program by reviewing three enhanced artifacts that represent software engineering, algorithms and data structures, and database technologies. These artifacts highlight the technical skills and professional competencies I developed during my coursework.

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25 – 50 seconds

Here's our agenda for today. We'll examine three artifacts spanning Software Design and Engineering, Algorithms and Data Structures, and Databases. I'll discuss the original functionality of each artifact, identify areas for improvement, and explain the enhancements I implemented. I'll conclude with a walkthrough of my professional ePortfolio and discuss how these artifacts demonstrate achievement of the program outcomes.

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50 – 1:08

These three artifacts were selected because they represent different areas of computer science and provide opportunities to demonstrate enhancements in security, performance, maintainability, and user experience. Together, they illustrate how I applied knowledge gained throughout the program to improve existing software solutions.

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1:08 – 1:34

Our first artifact is an Android Weight Tracking Application developed in CS 360. This mobile application provides a personal health management solution that allows users to create accounts, track their weight, set goals, and receive SMS notifications when they achieve target milestones. The application uses SQLite for local data storage and includes user authentication as well as full CRUD functionality.

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1:34 – 1:51

The application begins with a login system and transitions to the main tracking interface where users can record weight entries, view historical data, and monitor progress toward their goals. These features provide users with a simple and effective way to manage personal fitness information.

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1:51 – 2:09

One of the most significant issues in the original application was a critical security vulnerability. User passwords were stored in plain text within the SQLite database. This approach created a serious risk because anyone who gained access to the database could view user credentials and potentially compromise user accounts.

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2:09 – 2:30

To address this issue, I implemented secure password hashing using SHA-256 with salt before storing passwords in the database. I also added comprehensive error handling using try-catch blocks throughout the application. These enhancements significantly improved security by protecting user credentials even if the database were compromised.

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2:30 – 2:49

Another limitation of the original application was its architecture. User interface logic, database operations, and business logic were all contained within the same activity classes. This violated the principle of separation of concerns and made the application more difficult to maintain, test, and expand.

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2:49 – 3:10

My enhanced version separates responsibilities into distinct components using a Model-View-Controller style architecture. User interface operations, business logic, and database interactions are clearly separated. I also implemented additional input validation and error handling, resulting in code that is more maintainable, scalable, and reliable.

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3:10 – 3:29

Through these enhancements, I gained experience applying secure coding practices, architectural design principles, and software engineering techniques that improve both maintainability and user security. This artifact demonstrates how software engineering decisions can directly impact application quality and user trust.

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3:29 – 3:49

Our second artifact is a Pirate Intelligent Agent developed in CS 370. This project uses reinforcement learning and deep Q-learning to navigate a treasure-hunting environment. The intelligent agent learns optimal actions through experience replay and neural network training while balancing exploration and exploitation during the learning process.

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3:49 – 4:08

The original implementation relied on random experience replay, which treated every learning experience equally. This approach was inefficient because some experiences provide more valuable learning opportunities than others. As a result, the model required additional time to converge on an optimal solution.

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To improve performance, I implemented prioritized experience replay using Python's `heapq` module. By calculating Temporal Difference errors, the system prioritizes experiences that contain the most valuable learning information. This allows the agent to focus on important experiences and learn more efficiently.

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The enhancement produced measurable results. Training time decreased from approximately 10.36 minutes to 7.2 minutes, representing a 28.8 percent improvement in convergence speed. This demonstrated how selecting an appropriate data structure can significantly improve the efficiency of a machine learning system.

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This enhancement reinforced the importance of selecting appropriate algorithms and data structures to improve efficiency and overall system performance. It also demonstrated how theoretical computer science concepts can produce meaningful improvements in real-world applications.

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Our third artifact is the Animal Shelter Dashboard from CS 340. This Python and Dash web application uses MongoDB to help rescue organizations identify animals suitable for specialized training programs. The application includes interactive filtering, data visualization, and location-based mapping features.

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The original implementation retrieved entire database documents even when only a few fields were required. This increased resource consumption, transferred unnecessary data, and placed additional processing demands on the application.

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To improve efficiency, I implemented field projection to retrieve only the necessary data. I also added comprehensive database error handling, optimized large dataset displays, and centralized database operations into a dedicated module. These changes improved both maintainability and application performance.

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I further enhanced the user interface by providing real-time summaries and performance indicators that help users better understand query results. Users can quickly see how many records match their search criteria while maintaining efficient performance with larger datasets.

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This artifact provided valuable experience in database optimization, error handling, and improving application performance through more efficient data retrieval techniques. These enhancements resulted in a more effective and user-friendly solution.

Slide 21      6:26 – 6:45

Collectively, these artifacts demonstrate achievement of the program outcomes. They show my ability to solve computing problems, communicate technical concepts, collaborate through code review and documentation processes, apply innovative solutions, and incorporate security best practices throughout the software development lifecycle.

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To showcase this work professionally, I created an ePortfolio that documents each artifact and enhancement. The portfolio provides a centralized location for demonstrating technical skills, project growth, and achievement of program outcomes.

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The portfolio begins with a professional self-assessment that reflects on my growth throughout the Computer Science program. It explains how the selected artifacts demonstrate both technical development and readiness for professional software development roles.

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Each artifact includes detailed enhancement descriptions, technical explanations, demonstrated skills, and alignments with course outcomes. The portfolio also includes links to both the original and enhanced repositories, allowing viewers to examine the complete development process. The intelligent agent project prominently highlights the 28.8 percent performance improvement as measurable evidence of technical achievement.

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The portfolio concludes with a comprehensive overview of technical skills developed throughout the program along with professional contact information. Together, these elements create a complete professional package that showcases both technical competence and career readiness.

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In summary, these three artifacts illustrate my growth throughout the Computer Science program. Each enhancement addressed a specific limitation in the original artifact while providing opportunities to strengthen my skills in software engineering, algorithms and data structures, and database technologies.

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Together, these enhancements provide evidence of my readiness to contribute to professional software development teams and solve complex technical challenges. They demonstrate not only technical ability but also a commitment to security, maintainability, efficiency, and continuous improvement.

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Thank you for your time and attention. This code review reflects the knowledge, skills, and experience I have gained throughout the Computer Science program. I appreciate the opportunity to share these enhancements and demonstrate my growth as a software developer. I welcome any questions regarding the artifacts, enhancement decisions, or future development plans.